

Data Mining Applications

- ❑ Data Mining for Sentiment and Opinion
- ❑ Truth Discovery and Misinformation Identification
- ❑ Information and Disease Propagation
- ❑ Productivity and Team Science 

Productivity and Team Science

- ❑ Team components: (1) team leader and members, (2) the environment in which team members collaborate, (3) the task on which the team works.
- ❑ Data mining research questions:
 - ❑ Team performance characterization: how to reveal key characteristic patterns that differentiate a high-performing team from a struggling one?
 - ❑ Team performance prediction: how to forecast the performance of the team before or soon after the start of the task?
 - ❑ Team performance optimization: how to further enhance the team performance by adjusting the team composition?
 - ❑ Team performance explanation: how to interpret the team performance prediction and optimization results intuitively?

Productivity and Team Science

□ Team performance characterization:

□ Challenges:

- Team members possess different type of skills and social connectivity.
- The nature of tasks varies depends on the application scenarios.
- Environments are large, noisy, incomplete and volatile to dynamics.

- Treat as outcome of collective intelligence from both virtual teams and teams with face-to-face interactions.

□ Team performance prediction:

□ Challenges :

- Identify crucial features.
- Model the correlation between features and the team performance.
- Encode the dynamics of team evolution.

- Predict the impact of contents that team produces at a finer granularity.

Productivity and Team Science

❑ Team performance optimization:

- ❑ Optimize the team composition to maximize the performance.
- ❑ Aspects considered by the replacement algorithms:
 - ❑ Skills of the team members.
 - ❑ Network connectivity (the environment).
- ❑ Generalizations: team shrinkage, team expansion, team conflict resolution, real-time team optimization, etc.

❑ Team performance explanation:

- ❑ Limited literature exists for intuitive explanations of team performance/algorithms.
- ❑ Use influence function to identify key elements in teams for interpretation.
- ❑ Interpret from multiple levels.